Lecture 7

January 29, 2026

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Let's Start Simple

- As you all very well know, calculating the simple return is:

$$R = (P_1 - P_0 + D)/P_0$$

OR

$$\frac{\text{Simple return = Ending price of asset – beginning price of asset + Cash Dividened}}{\text{Beginning Price of Asset}}$$

- Retrospective look, but looking forward there is considerable uncertainty

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Expected Return

- We can think about future returns by predicting what outcomes may occur and at what probabilities these outcomes occur.
- This approach is called scenario analysis.
- How do we characterize this probability distribution?
- First, we measure the expected, or mean, rate of return:

$$E(R) = \sum_s p(s)\, R(s)$$

- Where p(s) is the probability of scenario and r(s) return of that scenario
- Expected return is the probability-weighted average across states
- Represents the investor's forecast, not a realized outcome

How To Quantify Uncertainty?

- While we may have an expected return, the return in any actual scenario could be more or less than the expected return.
- We measure this uncertainty using deviations of actual returns around the mean return.
- In particular, we use squared deviations from the expected returns to calculate the variance:

$$\mathrm{Var}(R) = \sigma^2 = \sum_s p(s)\,[R(s) - E(R)]^2$$

- Variance measures total dispersion of returns

Standard Deviation

- Use of squared deviations changes units.
- To get back to original units, we calculate the Standard Deviation.
- This is done by taking the square root of variance:
- $\sigma = \sqrt{\sigma^2}$
- Not a measure of good or bad, just treats both a deviations from the mean.
- This does mean that, as long as probability distribution is approximately symmetric, Standard Deviation is a reasonable measure of risk

Excess Returns and Risk Premium

- Excess return is defined relative to the risk-free rate R_f

$$r = R - R_f$$

- Risk Premium is the expected excess return:

$$E(r) = E(R) - R_f$$

- Interpreted as compensation for bearing risk
- Central object in asset pricing models
- This is where risk aversion plays in

Risk Aversion

First Game: Choose between $50 for certain and an equally-weighted gamble of either $ 100 or $ 0

Option A

$ 50 For certain

Option B

$ 0 If heads

$100 If tails

What happens if we increase the stakes a little?

Option A

$ 500 For certain

Option B

$ 0 If heads

$1,000 If tails

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Sharpe Ratio

- The Sharpe ratio measures risk-adjusted return—how much excess return you're getting for each unit of risk you take.
- Ratio of risk premium to Standard Deviation of Excess Return:

$$S = E(r)/\sigma$$

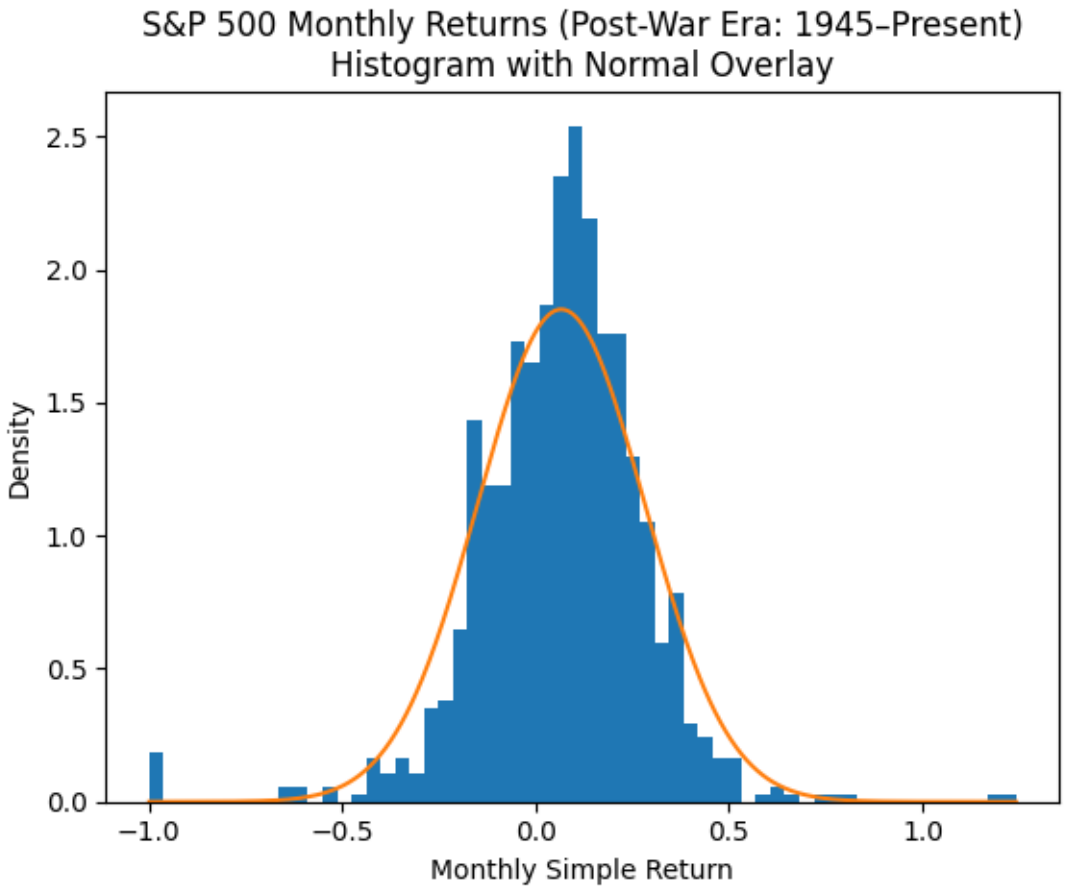
- Note for clarity: E(r) is risk premium, not expected return, E(R)
- Higher is better: more return per unit of risk
  - ≈ 0: returns roughly equal to risk-free rate
  - < 0: underperforming the risk-free rate
  - ~1.0: decent~
  - 2.0: very good≥
  - 3.0: exceptional (rare and often unstable)

Sharpe Ratio (con.)

- Why it’s useful
  - Compares portfolios with different risk levels
  - Helps evaluate whether higher returns are justified by higher volatility
  - Common in hedge funds, asset allocation, and factor analysis
- Limitations
  - Assumes returns are normally distributed (coming to the shortly)
  - Penalizes upside and downside volatility equally
  - Can be misleading for assets with skewed or fat-tailed returns (options, crypto, private credit)

What About the Distribution Of Historical Returns?

- See Figure showing distribution of monthly rate of return over post-WW2.
- Unlike fixed income, the distribution of a broad market index seems fairly described by a normal distribution.



Data: S&P 500 monthly price series (Shiller-style historical dataset).

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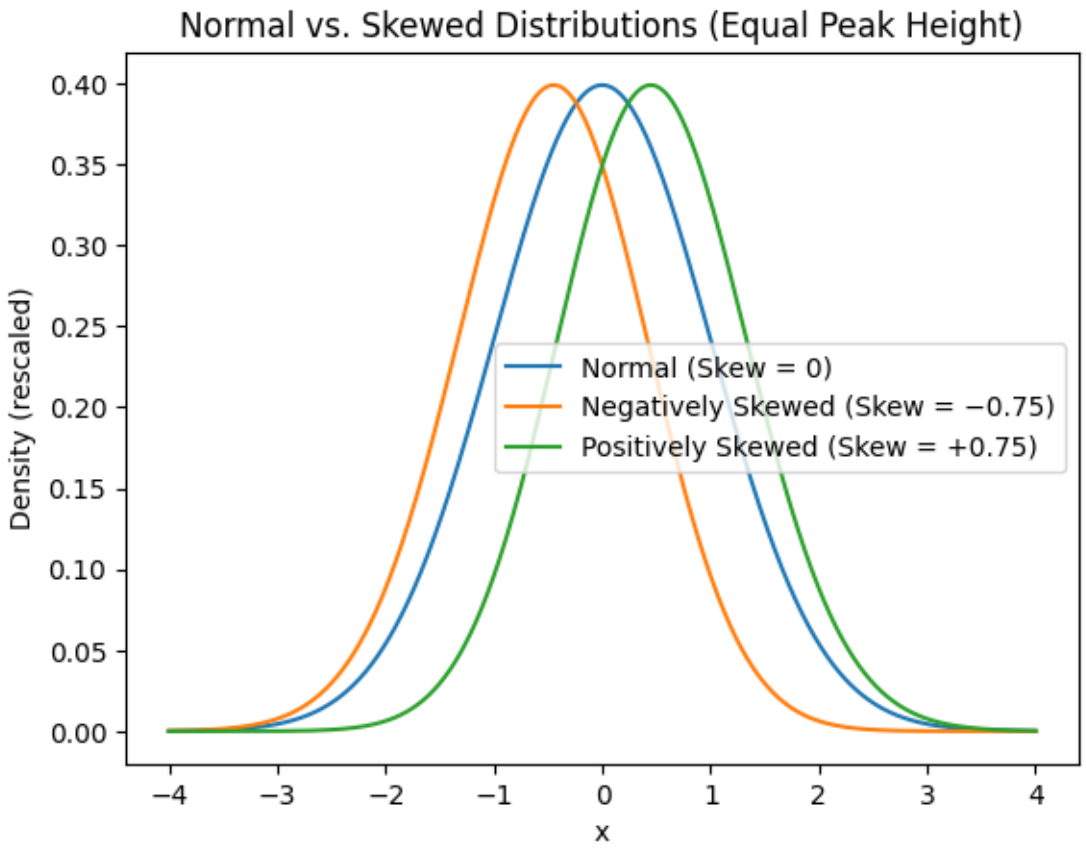
Why Normality Is Useful

- Common assumption that market returns are at least approximately normally distributed.
- Investment management would be much harder without this assumption. Why?
  - Symmetry – any positive deviation above mean = negative deviation below of the same magnitude
  - When assets with normally distributed returns are mixed to construct a portfolio, the portfolio return also is normally distributed (CLT)
  - Mean–variance utility is exact under normal returns (only two parameters)
  - When normally distributed, the statistical relationship between returns can be summarized with a single correlation coefficient

Departures From Normality

- Normality is a strong assumption often violated empirically
- Skewness measures asymmetry in a distribution.
  - A symmetric distribution has skewness ≈ 0
  - Negative skewness: longer or heavier left tail
  - Positive skewness: longer or heavier right tail
- Skewness reflects where extreme outcomes occur, not overall volatility.
- In finance, negative skew is often more concerning because large losses occur more frequently than large gains.

$$\text{Skewness} = \mathbb{E}\left[\left(\frac{R - \bar{R}}{\sigma_R}\right)^3\right]$$



Note that negative deviations raise to the 3rd power remain negative

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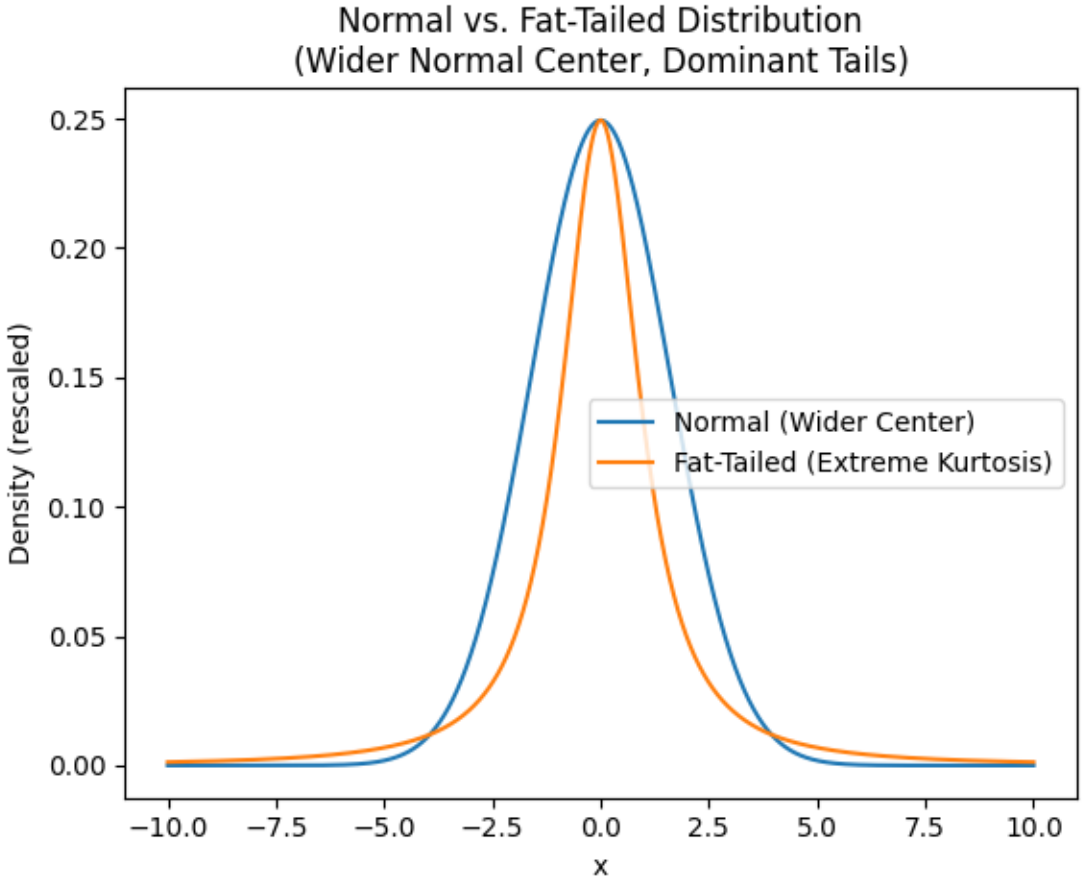
Kurtosis

- Kurtosis measures tail thickness, not peak height.
- High kurtosis (fat tails) means:
  - more extreme outcomes
  - higher probability of rare events
- High kurtosis means risk is concentrated in the tails, even if the center looks well-behaved.
- Because the normal distribution has kurtosis 3, we usually report excess kurtosis:

$$\text{Excess Kurtosis} = \mathbb{E}\left[\left(\frac{R-\bar{R}}{\sigma_R}\right)^4\right] - 3$$



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Lower Partial Standard Deviation (LPSD)

- When asymmetry enters, case for considering negative returns separately
- As we could choose risk-free portfolio over risky portfolio, consider deviations from target other than mean return (risk-free rate)
- Use LPSD – measures downside risk only
- Let R_t be returns and R_T a target:

$$\text{LPSD} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}\min(R_t - R_T,\, 0)^2}$$

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Sortino Ratio

- Risk-adjusted performance using downside risk instead of total volatility.

$$\text{Sortino Ratio} = \frac{\bar{R} - R_T}{\text{LPSD}}$$

Metric	Risk measure	Penalizes upside?
Sharpe Ratio	Total standard deviation	Yes
Sortino Ratio	Downside deviation only	No

- Not all volatility is bad — the Sortino ratio recognizes this.

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Sortino Ratio Example

- Assumptions:
  - Risk-free (or target) return = 2%
  - Portfolio annual returns (%):

Year	Fund A	Fund B
1	12%	5%
2	9%	4%
3	−6%	3%
4	11%	4%
5	−4%	1%

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Sortino Ratio Example (con.)

- Fund A Calculations
  - Average return = (12 + 9 − 6 + 11 − 4) / 5 = 4.4%
  - Excess return = 4.4% − 2% = 2.4%
  - Downside returns (< 2%): −6%, −4%
  - Deviations from target: −8%, −6%
  - Downside deviation = √[(64 + 36) / 2] ≈ 7.1%

$$\text{Sortino}_A = \frac{2.4\%}{7.1\%} \approx \mathbf{0.34}$$

- Fund B Calculations
  - Average return = (5 + 4 + 3 + 4 + 1) / 5 = 3.4%
  - Excess return = 3.4% − 2% = 1.4%
  - Downside returns (< 2%): 1%
  - Deviation from target: −1%
  - Downside deviation = √[(1) / 1] = 1.0%

$$\text{Sortino}_B = \frac{1.4\%}{1.0\%} = \mathbf{1.40}$$

Take A Look Back

- So far, we have been thinking in terms of forward-looking scenario analysis:
  - Identify relevant future scenarios
  - Assign probabilities and expected returns to each scenario
  - Compute:
    - Expected return (risk premium)
    - Standard deviation (risk)
- However, we can analyze historical returns
  - Based on time series of realized asset returns
  - Actual outcomes may differ from investors' original probability assessments
  - Allows only inference of:
    - Underlying return distribution, or
    - Estimated expected return and standard deviation

Historical Returns

- What if we considered each of those returns a scenario with equal probability?
- In finance, we often estimate the historic mean return on a particular asset as the average of all observed historic returns:

$$\bar{R} = \frac{1}{n}\sum_{s=1}^{n} r(s)$$

$r(s)$ = realized return in period s

n = number of observations

- In finance, we calculate the standard deviation of a historic series of daily returns:

$$\sigma = \sqrt{\frac{1}{n-1}\sum_{s=1}^{n}\left(r(s) - \bar{R}\right)^2}$$

- We often use the term Volatility in place of Standard Deviation

  σ = Volatility

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Averaging Returns: A Subtle but Critical Distinction

- Investment returns vary over time - how we average those returns depends on the question we are asking.
- Two commonly used averages:
  - Arithmetic average return
  - Geometric average return
- These:
  - measure different economic concepts
  - answer different questions
  - can lead to very different conclusions about performance
- The way returns are averaged matters just as much as the returns themselves.
- We will see how volatility creates a gap between arithmetic and geometric average returns.

Arithmetic Average Return

- We just saw this – it is the simple average of periodic returns:

$$\bar{r}_{\text{arith}} = \frac{1}{n}\sum_{s=1}^{n} r(s)$$

- Expected return in a single period
- Appropriate for:
  - forecasting next-period returns
  - asset pricing and mean–variance analysis
- Does not account for compounding

Geometric Average Return

- The compound average growth rate over time:

$$\bar{r}_{\text{geo}} = \left(\prod_{s=1}^{n} \left(1 + r(s)\right) \right)^{1/n} - 1$$

- Average return actually earned over time
- Correct measure for long-horizon performance
- Fully incorporates compounding

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Why They Are Different

- When returns vary:
  - Losses reduce wealth more than equal gains increase it
  - Volatility lowers compound growth
- As a result:

$$\bar{r}_{\text{geo}} \leq \bar{r}_{\text{arith}}$$

- Equality holds only if $r(s)$ is constant across periods.

Simple Example

- Returns:
  - $r(1)$=+50%
  - $r(2)$=−50%
- Arithmetic average: $\bar{r}_{\text{arith}} = \frac{0.50 + (-0.50)}{2} = 0\%$
- Geometric average: $\bar{r}_{\text{geo}} = \left(1.5 \times 0.5\right)^{1/2} - 1 = -13.4\%$
- This is known as volatility decay

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Connection to Volatility

- For moderate returns: $$\bar{r}_{\text{geo}} \approx \bar{r}_{\text{arith}} - \frac{1}{2}\sigma^2$$

- where σ^2 is the variance of $r(s)$.
- Higher volatility → larger gap between averages
- Volatility is a drag on growth

Question	Use
Expected one-period return	Arithmetic
Long-run investor experience	Geometric
Asset pricing models	Arithmetic
Performance reporting	Geometric

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Volatility Over Time and Scaling Laws

- Let returns be measured over discrete time periods (e.g., days, months)
- Let σ_1^2 denote the variance of one-period returns
- Let T denote the number of periods in the investment horizon
- Let one-period returns have variance σ_1^2
- Under independent and identically distributed (IID) returns:

$$\mathrm{Var}(R_T) = T\,\sigma_1^2$$

- Hence volatility scales as: $\sigma_T = \sqrt{T}\,\sigma_1$
- This is the square-root-of-time rule used in practice
- Assumption requires independence and finite variance

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Specific Case: Annualizing Volatility

- To convert periodic volatility to annualized volatility, multiply by the square root of the number of periods per year, e.g.,

$$\sigma_{annual} = \sqrt{12}\ \sigma_{monthly}$$
$$= \sqrt{52}\ \sigma_{weekly}$$

- The periodicity of volatility is based on the time interval between the original data observations, not the overall length of the time series
- It doesn't matter how many observations in the series
- The standard deviation of a series of daily returns must be multiplied by $252^{1/2}$ to convert to σ_{annual} . Why?

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Violations of IID Assumptions and Time-Varying Volatility

- IID assumption requires returns to be independent and identically distributed over time
- Empirically, financial returns violate this assumption in systematic ways
- Volatility clustering: large (small) returns tend to be followed by large (small) returns
- Formally, returns exhibit conditional heteroskedasticity: $\mathrm{Var}(R_t \mid \mathcal{F}_{\{t-1\}}) \neq$ constant
- Models such as ARCH and GARCH capture time-varying conditional variance
- Implication: unconditional variance may be stable, but short-horizon risk is predictable
- Consequence for scaling: $\sqrt{T}$ rule may overstate or understate true long-horizon risk

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Normality and Long-Term Investing

- Central Limit Theorem: sums of IID returns converge to normality
- Convergence may be slow under fat tails and skewness
- Compounded wealth remains lognormal, not normal
- Long-horizon investors still face substantial downside risk
- CLT suggests aggregation toward normality
- Fat tails slow convergence
- Long-horizon risk remains substantial

Distribution of Long-Term Returns

- One-period returns are often modeled as approximately normal:

$$R_t \sim N(\mu, \sigma^2)$$

- However, wealth evolves through compounding, not addition:

$$W_T = W_0 \cdot \prod_{t=1}^{T} (1 + R_t)$$

- Taking logs yields additive dynamics:

$$\ln(W_T/W_0) = \sum_{t=1}^{T} \ln(1 + R_t)$$

- If log returns are IID and normal, log wealth is normal
- Hence terminal wealth is lognormally distributed
- Arithmetic mean governs one-period expected returns
- Geometric mean governs long-run wealth growth
- Volatility creates a wedge between arithmetic and geometric means

Long-Term Investing and the Appearance of Risk Reduction

- With compounding, wealth is bounded below by zero but unbounded above
- Over long horizons, the distribution of terminal wealth becomes right-skewed
- Probability of ending below initial wealth typically declines with horizon
- This creates the appearance that long-term investing is "less risky“

Interpreting Long-Horizon Risk: Probability vs Severity

- But… risk has two distinct dimensions:
  - Probability of adverse outcomes
  - Severity (magnitude) of adverse outcomes
- Long horizons tend to reduce the probability of losses
- Simultaneously, compounding increases dispersion of wealth outcomes
- Left-tail outcomes become rarer but more extreme
- Variance alone does not distinguish between probability and severity
- Long-term risk assessment requires examining the full distribution of wealth

Fair Games and Speculation

- Speculation is the assumption of meaningful risk to obtain a commensurate gain. What does this mean?
  - Is it commensurate to have an expected gain of the risk-free rate?
  - A risk-adverse investor would say no – remember out thought experiment from earlier
- A fair game is an investment with zero expected excess return
  - Risk-neutral investors are indifferent to fair games
  - Risk-averse investors may reject fair games due to uncertainty
- Speculation involves taking positions based on beliefs about mispricing
- Speculators may accept unfavorable odds if upside is sufficiently large
- Risk aversion explains why assets with identical expected returns are valued differently
- Distinguishes gambling from investing through expected utility, not expected value alone

Expected Utility Theory

- Risk-adverse investors reject a fair game in favor of certainty, but how to quantify the rate at which we are willing to trade off return against risk?
- Investors use a general framework of rational decision-making under uncertainty called expected utility.
  - People choose the option with the highest expected utility, not expected value.
  - Utility reflects subjective preferences toward outcomes.
  - Many functional forms and solution methods exist
  - EUT provides a unifying principle, not a single model

Mean–Variance Utility Approximation

- One widely used implementation of expected utility
- Approximates utility using expected return and risk

$$U = E(r) - \frac{1}{2}A\sigma^2$$

- Risk enters through return variance
- A summarizes risk aversion
- Standard in portfolio theory and asset pricing

Risk Preferences in Mean–Variance Utility

- Risk-averse: $A>0$
  - Disutility from volatility
  - Prefers lower risk for a given expected return
- Risk-neutral: $A=0$
  - Utility depends only on expected return
  - Indifferent to risk
- Risk-seeking (gambling): $A<0$
  - Volatility increases utility
  - Prefers riskier outcomes

Risk-Free Asset in Mean–Variance Utility

- Consider the mean–variance utility function: $U = E(R) - (1/2)A\sigma^2$
- A risk-free asset has known return R_f with certainty
- Therefore: $E(R_f) = R_f$ $Var(R_f) = \sigma_f^2 = 0$
- Substituting into mean–variance utility: $U_f = R_f - (1/2) \cdot A \cdot 0$
- Hence utility from the risk-free asset simplifies to: $U_f = R_f$
  - Risk-free asset delivers utility equal to its return
  - No utility penalty from risk
  - Serves as benchmark for evaluating risky portfolios

Portfolio of One Risky and One Risk-Free Asset

- Consider a portfolio with weight y in a risky asset and (1−y) in the risk-free asset
- Portfolio return: $R_p = yR + (1-y)R_f$
- Expected return: $E(R_p) = yE(R) + (1-y)R_f = R_f + y[E(R) - R_f]$
- Excess return form: $E(R_p - R_f) = yE(r)$
- Portfolio variance: $Var(R_p) = y^2Var(R) = y^2\sigma^2$ (since $Var(R_f) = 0$ and $Cov(R, R_f) = 0$)
- Portfolio volatility: $\sigma_p = y\sigma$
- Risk and return scale linearly with exposure y

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Capital Allocation Line (CAL)

- Combining the results for $E(R_p)$ and σ_p yields:

$$E(R_p) = R_f + \frac{E(r)}{\sigma}\sigma_p$$

- This equation defines the Capital Allocation Line (CAL)
- CAL is a straight line in mean–variance space
  - Intercept: R_f (return with zero risk)
  - Slope: $E(r)/\sigma$ (Sharpe ratio of the risky asset)
  - All portfolios formed from this risky asset and the risk-free asset lie on the CAL
- Investors differ only in their chosen point along the line, determined by risk aversion

Utility Maximization on the Capital Allocation Line

- Investor chooses portfolio to maximize mean–variance utility:

$$U = E(R_p) - (1/2)A\sigma_p^2$$

- Substitute CAL equation: $$E(R_p) = R_f + \frac{E(r)}{\sigma}\sigma_p$$

- Utility as a function of risk: $$U(\sigma_p) = R_f + \frac{E(r)}{\sigma}\sigma_p - \frac{A}{2}\sigma_p^2$$

- First-order condition: $$\frac{dU}{d\sigma_p} = \frac{E(r)}{\sigma} - A\sigma_p = 0$$

- Optimal risk exposure: $$\frac{E(r)}{\sigma} = A\sigma_p \quad \text{->} \quad \sigma_p^* = \frac{E(r)}{A\sigma}$$

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Example

- Risk-free rate: Rf=2%
- Risky asset (or risky portfolio): $E(R)$=8%, σ=20%
- Risk aversion (mean–variance): A=4
- So the expected excess return is: $E(r)$=$E(R)$−Rf=8%−2%=6%

1) Find the CAL : $E(R_p) = R_f + \frac{E(r)}{\sigma}\sigma_p$ -> $\frac{E(r)}{\sigma} = \frac{0.06}{0.20} = 0.30$

- So the CAL is: $E(R_p)$ = 0.02+0.30 σ_p
- So every +1% of portfolio volatility increases expected return by 0.30%

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Example (con.)

2) Find the optimal portfolio (utility maximization)

- Mean–variance utility: $U = E(R_p) - \frac{A}{2}\sigma_p^2$
- From Slide 40, we have: $\sigma_p^* = \frac{E(r)}{A\sigma} = \frac{0.06}{4 \cdot 0.20} = 0.075$
- So optimal portfolio volatility is 7.5%.
- Convert this to the risky weight y using $\sigma_p = y\sigma$: $y^* = \frac{\sigma_p^*}{\sigma} = \frac{0.075}{0.20} = 0.375$
- So the investor puts 37.5% in the risky asset and 62.5% in risk-free asset

Example (con.)

3) Compute $E(R_p*)$ and $U*$

- Expected return:
  - $E(R_p{}^*)$=R_f+$y^*E(r)$= 0.02 + 0.375 · (0.06) = 0.0425
  - So $E(R_p{}^*)$=4.25%
- Utility:
  - U^*=0.0425−(1/2) ·4·(0.075)^2= 0.0425−2 ·(0.005625) = 0.03125
  - So U^*=3.125%

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Let’s sensitize

- See where utility peaks:

y	σ_p	$E(R_p)$	$U = E(R_p) - \frac{A}{2}\sigma_p^2$
0.000	0.000	0.0200	0.02000
0.250	0.050	0.0350	0.03000
0.375	**0.075**	**0.0425**	**0.03125**
0.500	0.100	0.0500	0.03000
1.000	0.200	0.0800	~0.00000

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